

Umer Irfan

Lahore · Pakistan

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EDUCATION

Lahore University of Management Sciences

BSc Computer Science · Minor in Computer Engineering

Sept 2023 to July 2027

CS GPA **3.707/4.00** · CGPA **3.491/4.00**

Pak-Turk Maarif International Schools and Colleges

June 2023

A-Levels: 3A/A* · O-Levels: 10A*

RESEARCH INTERESTS

Embedded Systems · Hardware Security · Firmware Security · Computer Architecture · Operating Systems

HONOURS AND AWARDS

- **1st Place.** *Pakistan Semiconductor Summit Hackathon* 2026. RISC-V RTL verification with Sail and Verilator.
- **Best Project.** Intel ISEF Pakistan 2019.
- **Dean's Honour List.** LUMS. Three semesters.
- **Best in Category.** Teknofest 2020. Gaziantep, Turkey.
- **Team Pakistan.** *Young Physicists Tournament*, Beijing (2018).

PUBLICATIONS

Accepted Publications

- ***Motion-Coupled Sensing: Self-Powered State-Change Sensing*** Co-author · *IEEE MASS 2026*
 - Co-developed **batteryless hinge sensing** and LoRa reporting. Hinge motion powers each measurement and transmission without periodic polling or battery replacement.
 - Built ATmega328P hardware and a stepper-driven rig for 10,000 actuations. Cut a firmware loop from 520 to 7 cycles. Field deployment achieved **99.3% reliability** over 5,945 actuations at five sites.
- ***Kissan-Dost: Conversational IoT for Precision Agriculture*** Co-author · *IEEE DCOSS-IoT 2026*
 - Built a multi-hop ESP-NOW mesh and ESP32 gateway for soil and climate data. Delivered **Urdu WhatsApp text and voice guidance** without requiring farmers to interpret dashboards.
 - Five farmers in a pilot used the chatbot near-daily to guide irrigation during two 45-day site deployments.

Submitted Manuscript

- ***PARDA: On-Device Audio Privacy for Smart Glasses*** First author · *IEEE PerCom 2027*
 - Designed **context-aware audio redaction and voice replacement** to block cross-turn privacy inference.
 - Distilled a retrieval-aware 2B model on 717 CANDOR conversations. Tested quantized inference on RPi 5.
 - Across 143 held-out conversations, model-judged leakage fell from 0.502 to 0.378 while utility rose from 0.875 to 0.900. Pipeline weighted real-time factor was 0.77.

Manuscripts in Preparation

- ***Programmer-in-the-Middle: Firmware Implantation via Flash Programmers*** Joint first author · Pre-Release
 - Exposed a **firmware supply-chain gap**: programmers altered flashed code while passing host readback checks.
 - Modified USBasp firmware for AVR reset-vector hijacking and stub injection. Spoofed host readback checks.
 - Reverse-engineered ST-Link V2 firmware and tested custom programmer firmware across **three target boards**.
- ***On-Chip Behavioral Monitoring for Runtime Hardware Trojan Detection*** Joint first author · Pre-Release
 - Designed **transaction-level monitoring** to detect hardware Trojans missed by pre-deployment tests.
 - Built a trace-collection pipeline for expected and Trojan-affected execution across computational and application workloads. Developed a 1D CNN classifier.
 - Specified an **FPGA monitor architecture**. Planned generalization studies across processor and cache configurations with latency and resource evaluation.

Preprints

- ***WatchTower: Transformer-Based Open-Set Network Anomaly Detection*** Joint first author
 - Developed **open-set intrusion detection** for attacks absent from edge-device training data.
 - Used a transformer with 28 packet features and PROSER. Deployed INT8 inference via TVM on RPi 5.
 - Reached 3.82 ms inference latency and 93.66% classification accuracy on the deployed model.
- ***LLM-Enhanced Wearables for Health Guidance in LMICs*** Co-author · *arXiv:2602.08701*
 - Designed a **screenless wearable** with an ATmega328P, PPG, temperature and motion sensing, and BLE health feedback over WhatsApp. Unit cost was \$12.72.
 - A 20-participant study over 96 hours found mean health-data comprehension rose from 3.05 to 3.60/5.

SELECTED RESEARCH AND TECHNICAL PROJECTS

PixelPacker | FPGA-Accelerated Neural Image Compression

2025 to 2026 · [Project page](#)

- Built edge-cloud image compression for wildlife camera traps on bandwidth-constrained links. Exploited fixed backgrounds to avoid retransmitting unchanged content.
- Developed an **Artix-7 FPGA accelerator** with pipelined DSPs, FIFO buffering, and DDR memory access. Implemented background-aware latent differencing on ESP32.
- Distilled and INT8-quantized a depthwise-separable encoder from 28.4 MB to 800 kB. Reduced its memory footprint by approximately **36-fold** for edge deployment.

RIZZ-8 | Custom 8-Bit RISC Processor

2025 · [Project page](#)

- Independently designed and built an **8-bit Harvard CPU** from 74-series ICs on **eight custom PCBs** with a 16-bit address space.
- Defined a custom ISA with signed and unsigned arithmetic and bitwise operations.
- Wrote a Python assembler and loaded machine code into EEPROM. Executed custom assembly programs on the completed processor.

KeySiphon | Power-Rail Side-Channel Analysis

2025 · [Project page](#)

- Built an instrumented **RSA side-channel prototype** on ATmega328P. Isolated its supply and captured leakage traces through a shunt resistor.
- Trained a 1D CNN to distinguish exponent-bit operations with 93.10% validation accuracy. Automated inference from noisy traces.

CARDY | ESP8266-Based Interactive Challenge Platform

2023 · [Project page](#)

- Developed [ESP8266 firmware](#) and a custom PCB for an interactive hardware competition. Integrated an HTTP server with LittleFS and physical controls.
- Coordinated assembly and testing of **50 devices** used by 50 participating teams with real-time telemetry.

RESEARCH EXPERIENCE

Research Assistant | [SYSNET Lab](#) at LUMS

May 2024 to Sept 2026

[Dr. Naveed Anwar Bhatti](#)

- Built a laptop-controlled **voltage-trace replay bench** with function generators for repeatable CheckMate power tests.
- Built an aluminum **Peltier chamber** spanning 5 to 70°C to measure temperature effects on device performance.
- **Designed and hand-assembled a four-layer nRF52 wristband** for maternal-health temperature monitoring in rural Sindh. Targeted four months per charge.
- Co-developed a **lossless 16-bit thermal codec** with temporal and camera-noise predictors. Files averaged 15% smaller than FFV1 with no larger outputs in tested sets. Best case was one-third the size.

Research Assistant | [CHASE Lab](#) at LUMS

2025 to 2026

[Dr. Muhammad Ali Siddiqi](#)

- Developed FPGA datapaths for ML inference and characterized power side-channel leakage on embedded platforms.
- Assessed BLE power-exhaustion risk in a laboratory **pacemaker (IMD)** using HackRF and ESP32. Crafted packets extended wake time from about 2 to 10.8 seconds. Each attempt used 1.56 to 1.93 mJ.
- Contributed FPGA acceleration, programmer-firmware analysis and transaction-trace collection to PixelPacker, PITM and runtime Trojan monitoring.

TEACHING EXPERIENCE

EE320 Computer Organization & Assembly Language

Sole TA · 2026. Six quizzes. Seven homework sets.

AI624 AI on Edge Devices

TA · 2026. 82 graduate students.

ENGG100 Engineering Design Fundamentals

TA · 2024. 30 engineering freshmen.

CS370 Operating Systems

TA · 2026. 295 students. Kernel and shell assignments.

CS210 Data Structures

TA · 2025. Assignments and grading.

Diploma in Digital Health

Course Designer · 2026. Taught IoMT to 90+ medical professionals.

SELECTED GRADUATE COURSEWORK

Some graduate-level courses taken during my undergraduate degree:

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|---|-------------------------|
| • Internet of Things | • AI on Edge Devices |
| • Applied Cryptography | • Computer Architecture |
| • Topics in Computer and Network Security | • Distributed Systems |

PROFESSIONAL EXPERIENCE

- AI/ML Engineer | Call Agent Application

Aug 2024 to March 2025

 - Built a **streaming voice agent** using Node.js and WebSockets. Integrated WebRTC VAD with Whisper and GPT-4.
 - Cut response latency to **under four seconds** through audio buffering and pipeline optimization.
- DevOps Engineer | *U.n.I Social App*

May to July 2024

 - Built GitHub Actions workflows for containerized deployment and automated application delivery.
 - Configured EC2 infrastructure and NGINX reverse proxies for traffic and load management.
- React Native Engineer | Project X

Nov 2024 to June 2025

 - Developed an Android application for a smart security-camera startup.
 - Implemented **IoT device onboarding**, Firebase authentication, cloud messaging and backend API integration.

TECHNICAL SKILLS

Programming & HDL C · C++ · Python · x86, MIPS & RISC-V Assembly · Verilog · SystemVerilog Microcontrollers PIC16F876A/877A · ATmega328P · ATmega32U2/8U2 · ATtiny85/16 · STM32 · TM4C123GH6PM · nRF52832 · ESP32/ESP8266 FPGA & Architecture Basys 3 (Artix-7 XC7A35T-1CPG236C) · Nexys A7 (Artix-7 XC7A100T/XC7A50T) · RISC-V · CPU & ISA design · Sail · Verilator Embedded Systems SPI · I²C · UART · BLE · ESP-NOW · LoRa · FreeRTOS	Firmware Security USBasp · ST-Link V2 (SWD) · Ghidra · Binary rewriting · Readback spoofing Hardware Instrumentation Oscilloscope · Function generators · HackRF/SDR · Power-trace capture PCB & Mechanical Design KiCad · EasyEDA Pro · Eagle · JLCPCB · PCB etching · SolidWorks · Blender Edge Inference PyTorch · TensorFlow Lite Micro · TVM · INT8 quantization · Knowledge distillation · FedProx
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